

Coding & Solution

Date	02 July 2026
Team ID	xxxxxx
Project Name	Raising waters
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Bhargavi11111/rising-waters-project-.git
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn

Key Features Implemented	- Flood prediction using Machine Learning. - Comparison of Decision Tree, Random Forest, KNN, and XGBoost models. - XGBoost selected as the best model with 96.55% accuracy. - User-friendly Flask web application for flood risk prediction. - Weather parameter input and instant prediction results. - Prepared for IBM Cloud deployment.
Pending / Incomplete Features	- Real-time weather API integration. - SMS/Email alert notifications. - Interactive dashboard with live weather monitoring. - Multi-user authentication and authorization.
Setup / Run Instructions	- Install Python and required libraries (Flask, pandas, numpy, scikitlearn, xgboost). - Place the trained model (.pkl) and dataset in the project directory. - Run app.py using python app.py. - Open the Flask application in a web browser and enter weather parameters to obtain flood predictions.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project follows a modular architecture with separate components for data preprocessing, machine learning model training, prediction, and the Flask web interface. The XGBoost model achieved 96.55% accuracy, making it the most suitable model for flood prediction. The application is scalable and can be enhanced with real-time weather APIs, notification services, and IBM Cloud deployment for broader accessibility.